

Sales Analysis and Business Strategy Optimization

GROUP MEMBERS:

- Aarya Arban (05)
- Soham Babsheyte (06)
- Aarna Bafna (07)
- Dhruv Bharwada (08)

INTRODUCTION

This project aims to analyze sales data to uncover key trends, customer behavior, and performance patterns across products, regions, and time. By combining Exploratory Data Analysis (EDA), clustering, forecasting with Facebook Prophet, and visualization techniques, we derive actionable insights to support data-driven decision-making and business growth.

DATASET AND FEATURES

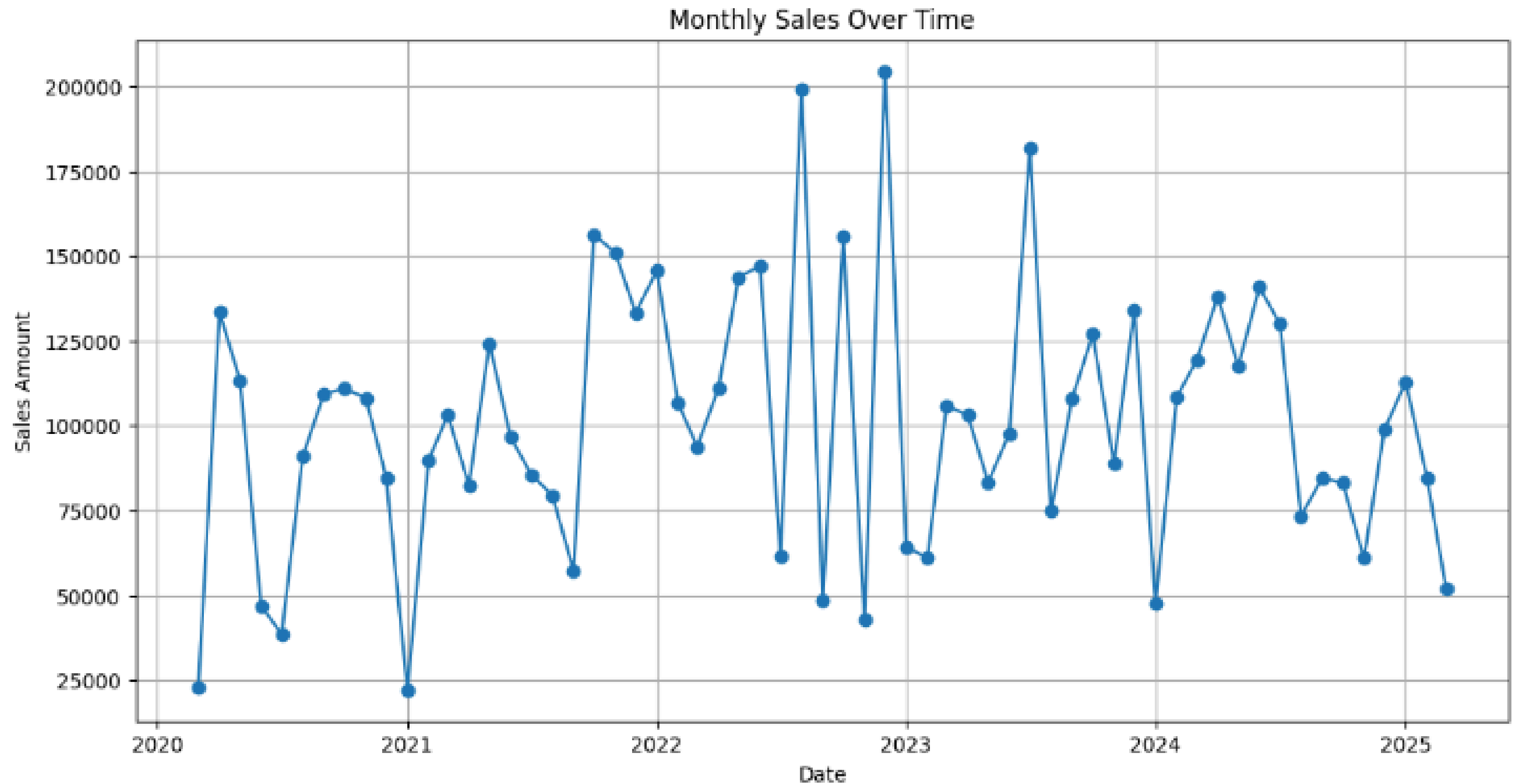
```
[3] sales_dataset.head()
```

	Order ID	Amount	Profit	Quantity	Category	Sub-Category	PaymentMode	Order Date	CustomerName	State	City	Year-Month
0	B-26776	9726	1275	5	Electronics	Electronic Games	UPI	2023-06-27	David Padilla	Florida	Miami	2023-06
1	B-26776	9726	1275	5	Electronics	Electronic Games	UPI	2024-12-27	Connor Morgan	Illinois	Chicago	2024-12
2	B-26776	9726	1275	5	Electronics	Electronic Games	UPI	2021-07-25	Robert Stone	New York	Buffalo	2021-07
3	B-26776	4975	1330	14	Electronics	Printers	UPI	2023-06-27	David Padilla	Florida	Miami	2023-06
4	B-26776	4975	1330	14	Electronics	Printers	UPI	2024-12-27	Connor Morgan	Illinois	Chicago	2024-12

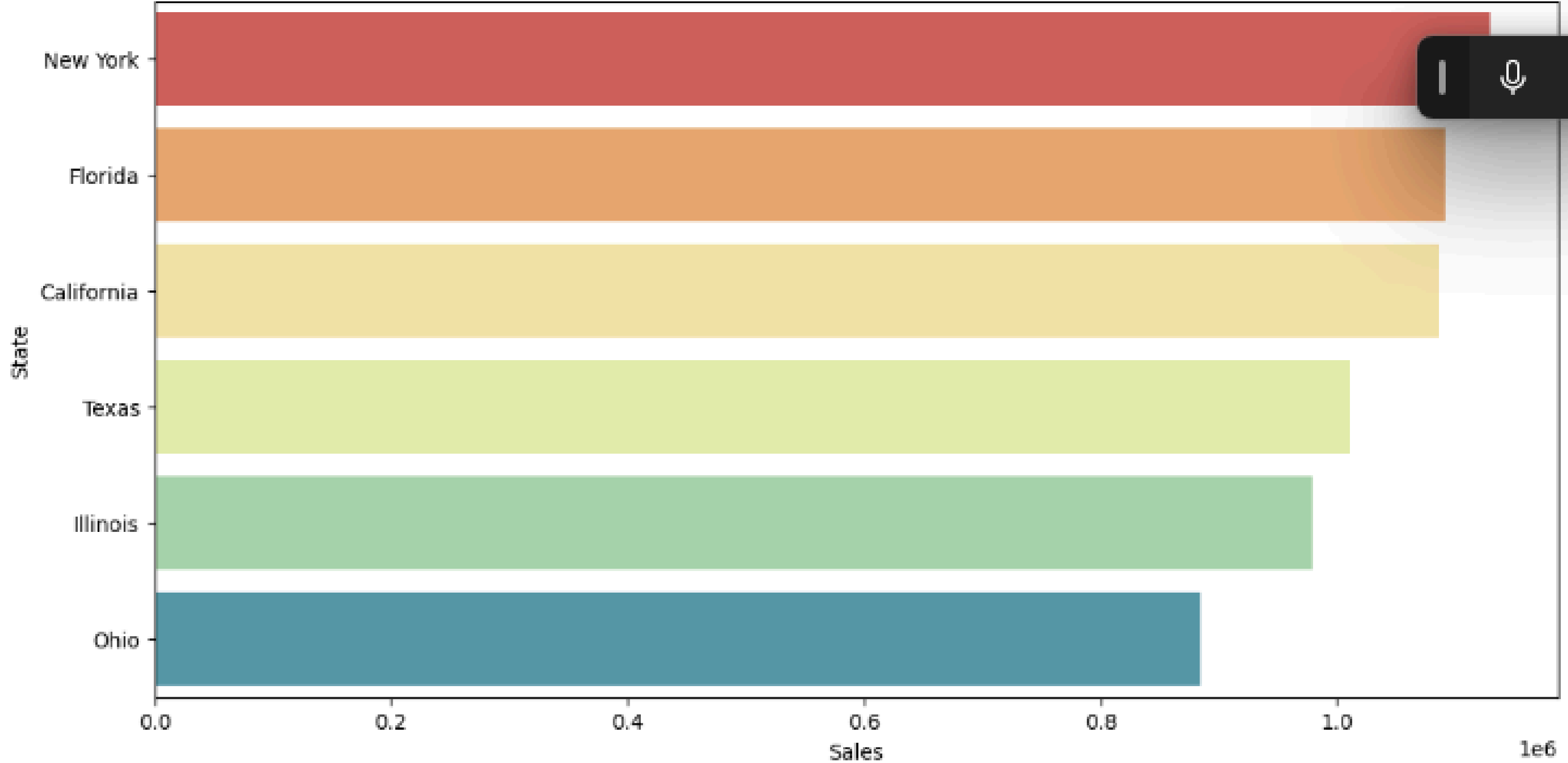
```
sales_dataset.describe()
```

	Amount	Profit	Quantity
count	1194.000000	1194.000000	1194.000000
mean	5178.089615	1348.992462	10.674204
std	2804.921955	1117.992573	5.777102
min	508.000000	50.000000	1.000000
25%	2799.000000	410.000000	6.000000
50%	5152.000000	1014.000000	11.000000
75%	7626.000000	2035.000000	16.000000
max	9992.000000	4930.000000	20.000000

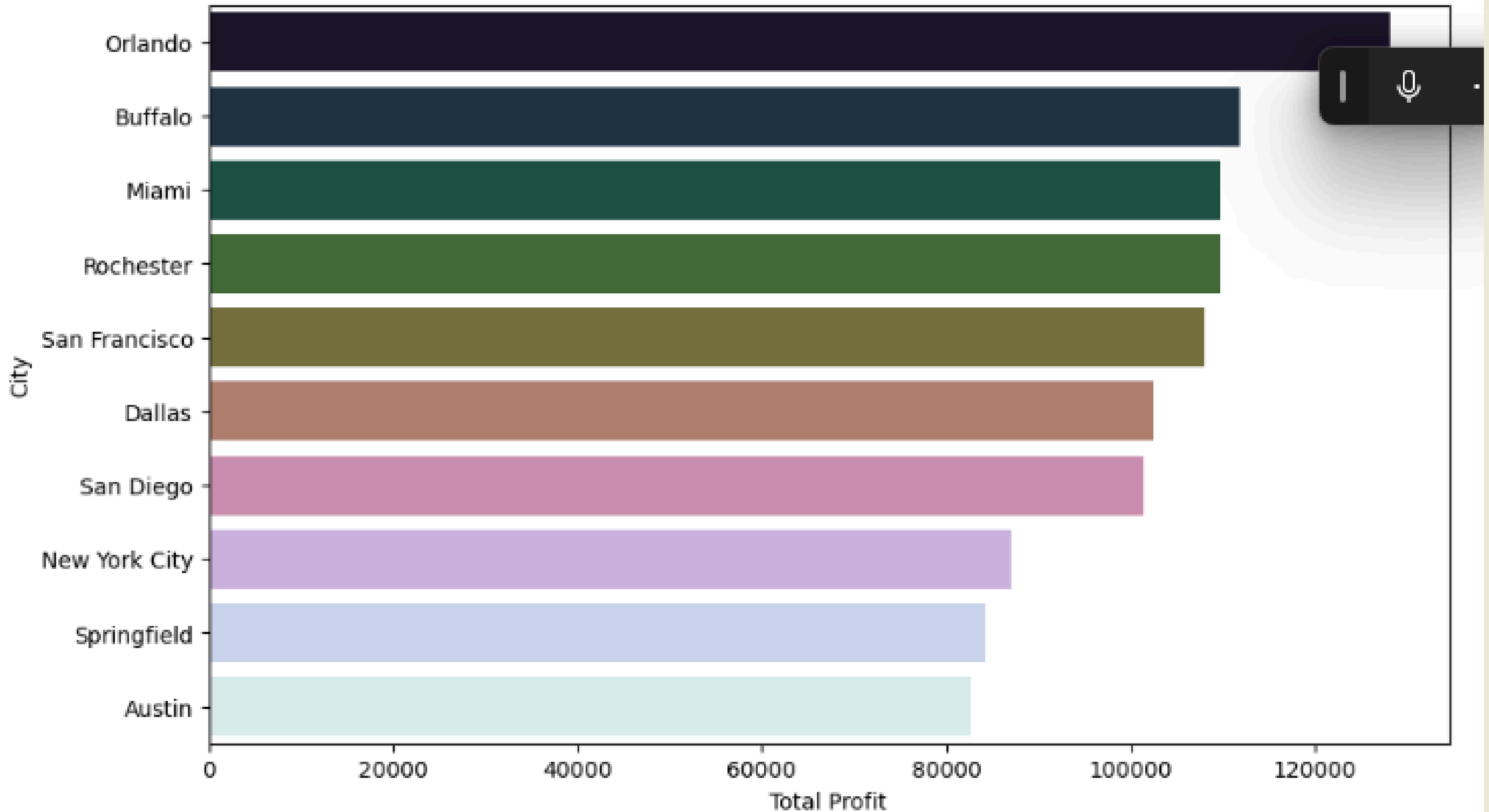
RESULTS

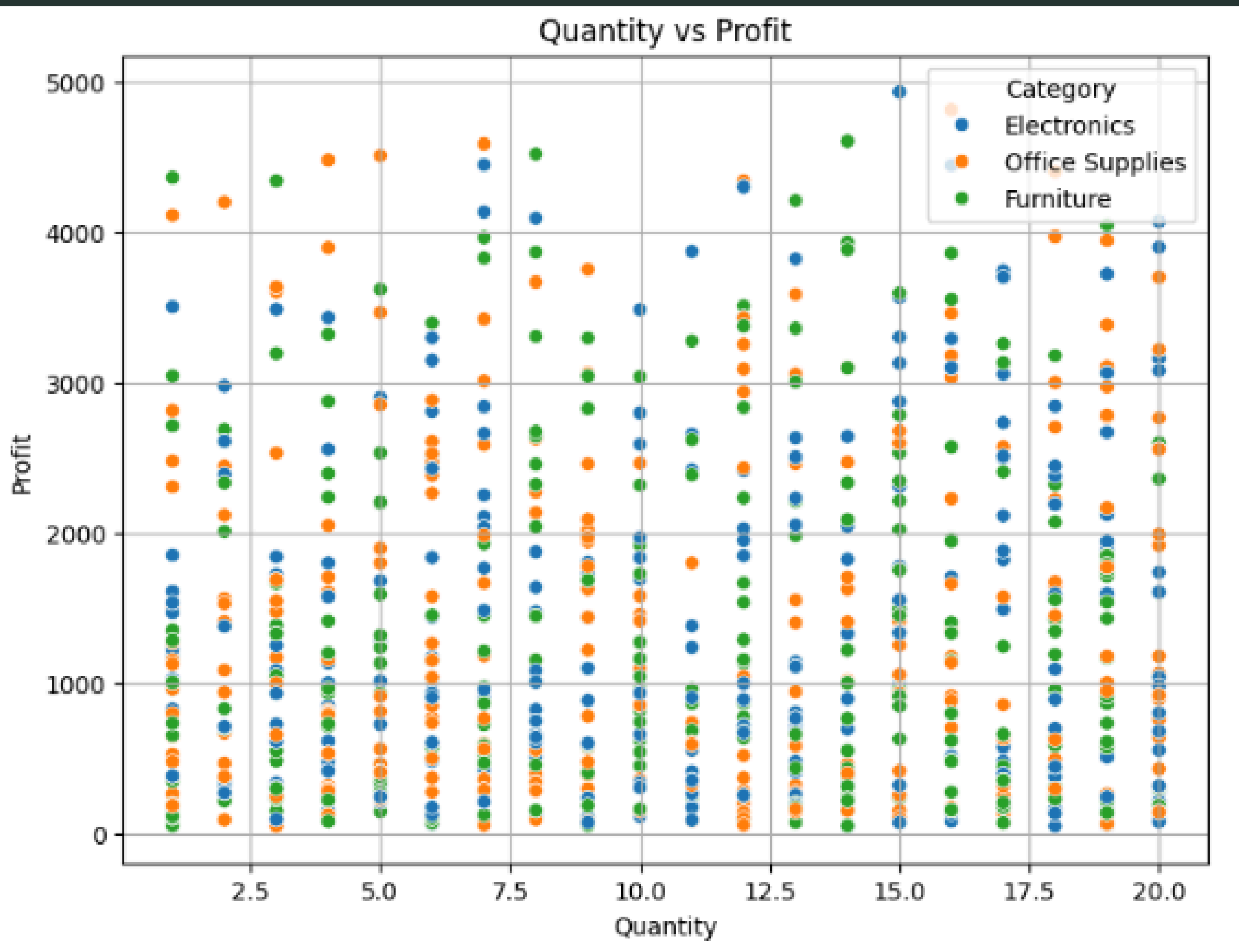


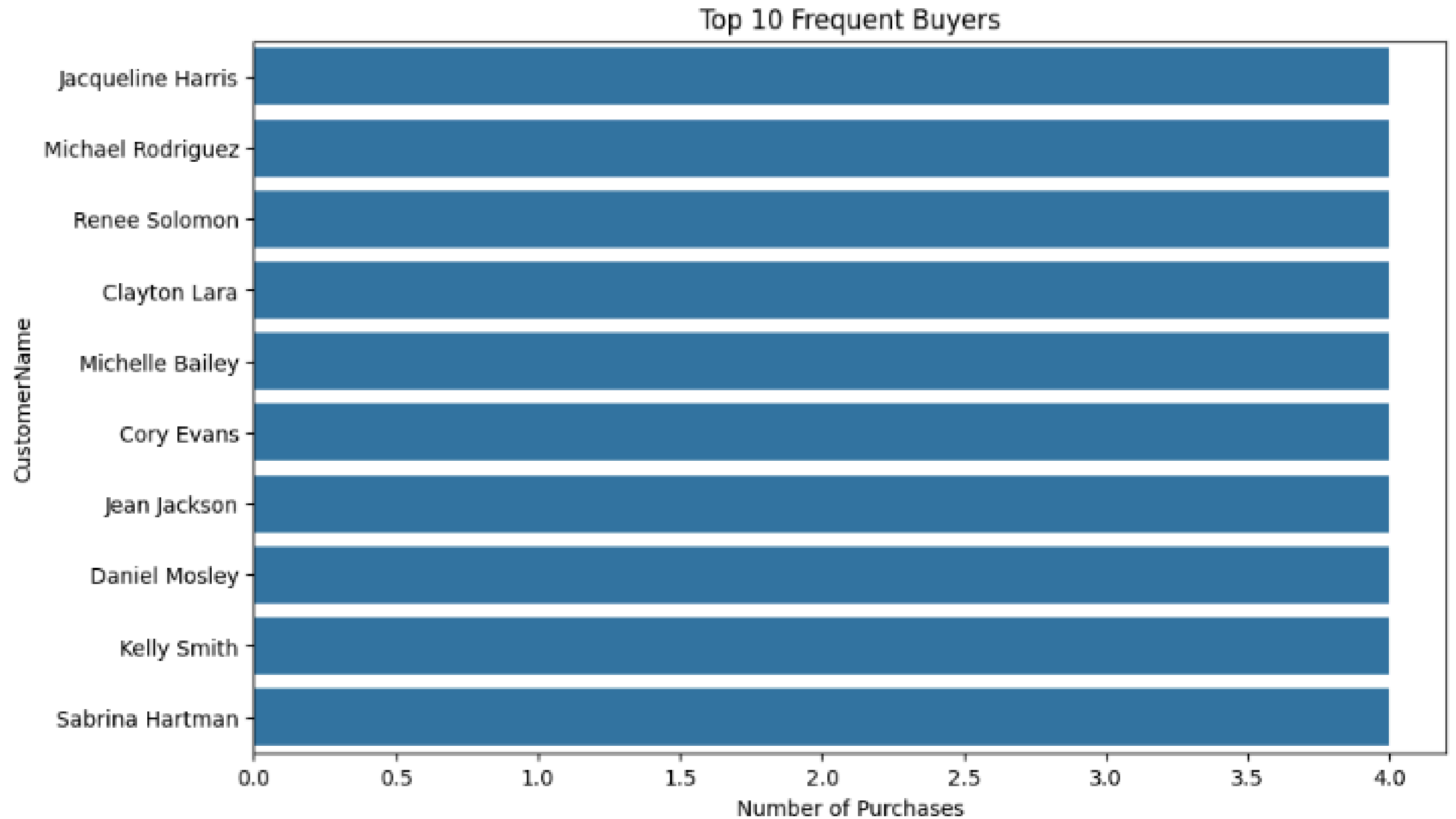
Sales by State



Top 10 Cities by Profit



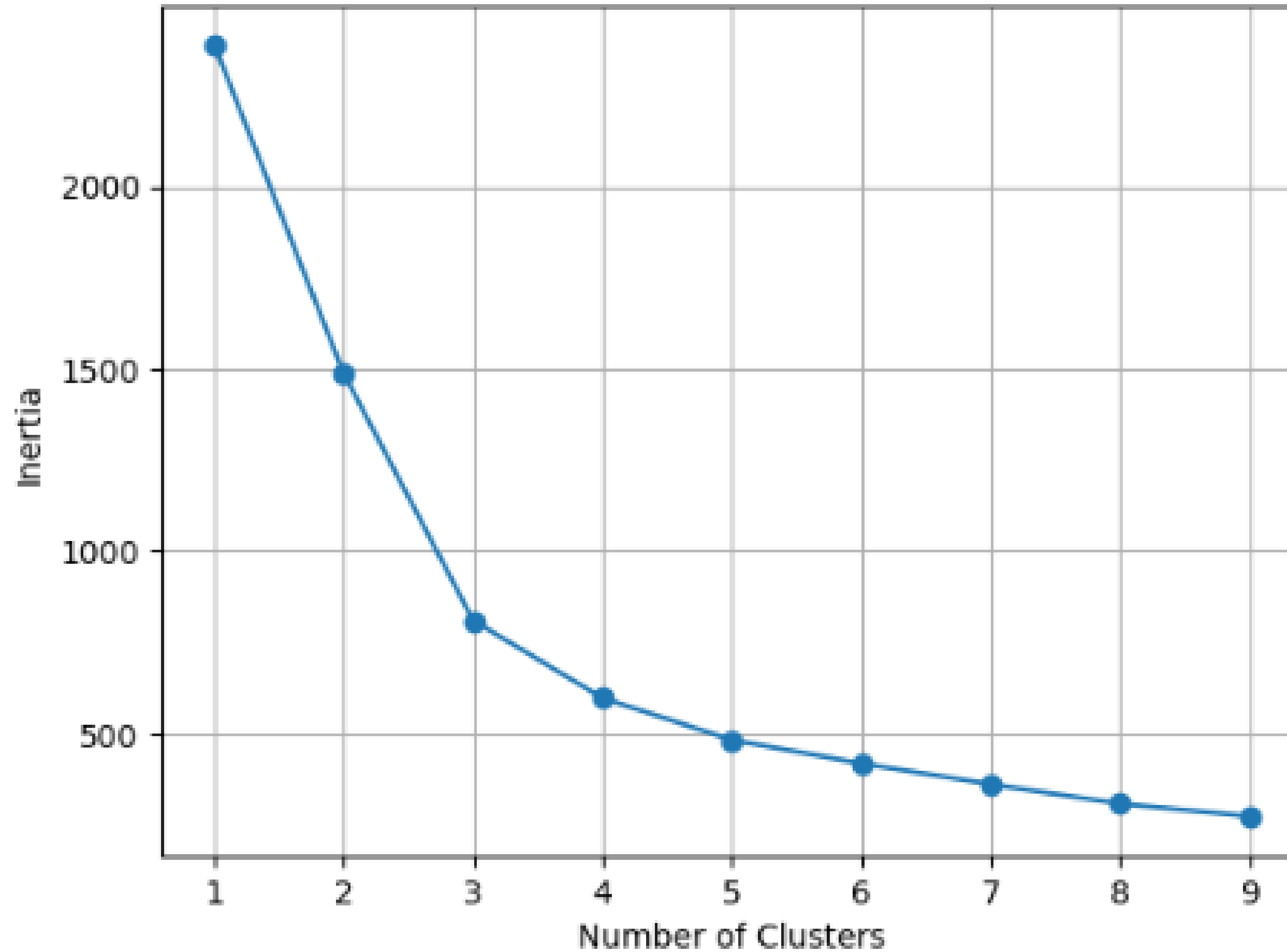




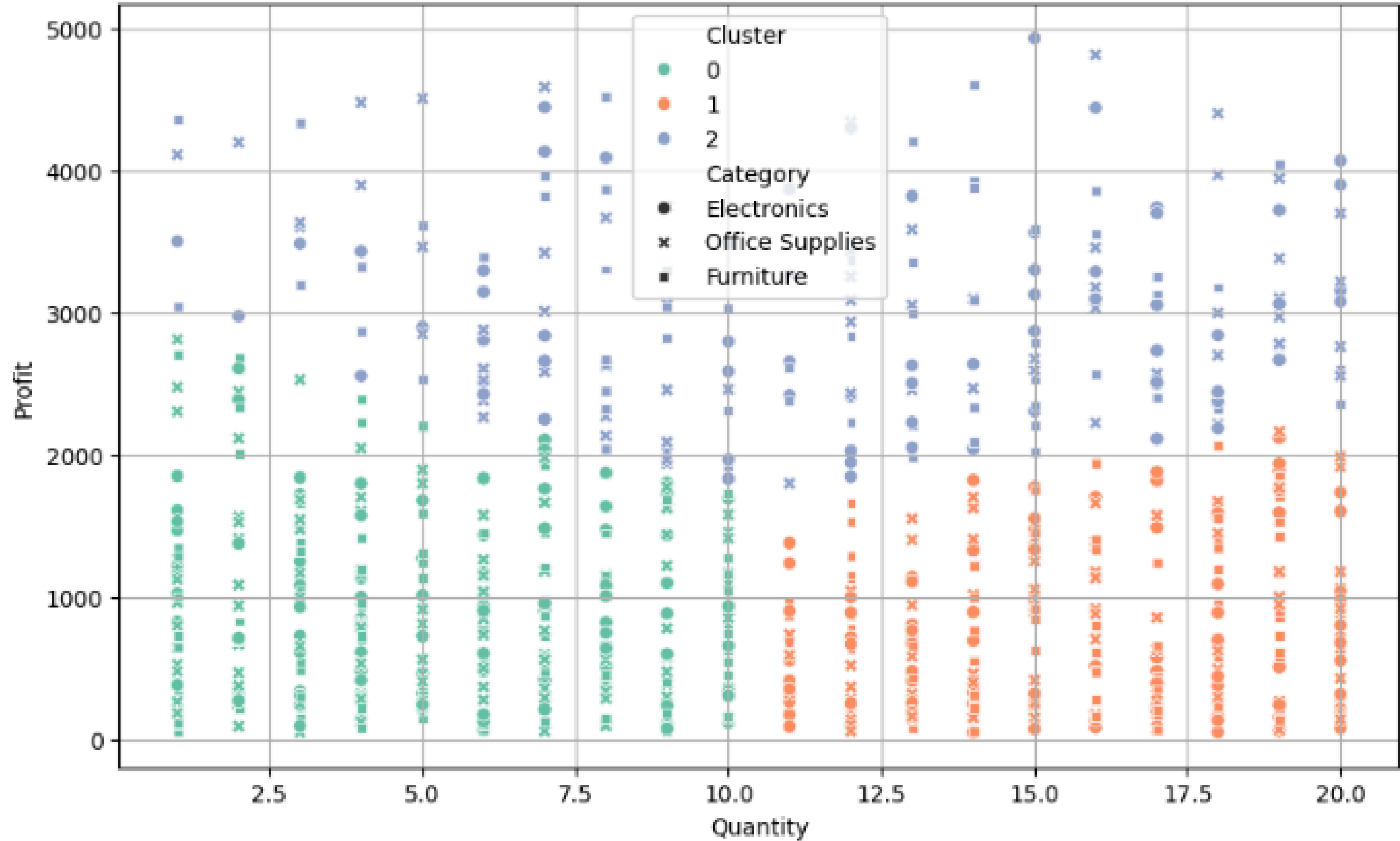
K - MEANS ALGORITHM

- K-Means clustering is a popular unsupervised machine learning algorithm.
- It automatically finds patterns by grouping similar data points together based on their features
- The algorithm divides the data into K distinct clusters, where K is a user-defined number. It works in the following steps:
 - Randomly initialize K centroids (one for each cluster).
 - Assign each data point to the nearest centroid based on distance (usually Euclidean distance).
 - Recalculate the centroids as the mean of all data points in a cluster.
 - Repeat steps 2 and 3 until the centroids no longer change significantly.
- The number of clusters, K, needs to be set manually. One commonly used method to find the optimal K is the Elbow Method, where we plot the Within-Cluster Sum of Squares (WCSS) against different values of K and choose the point where the rate of decrease sharply slows

Elbow Method for Optimal k

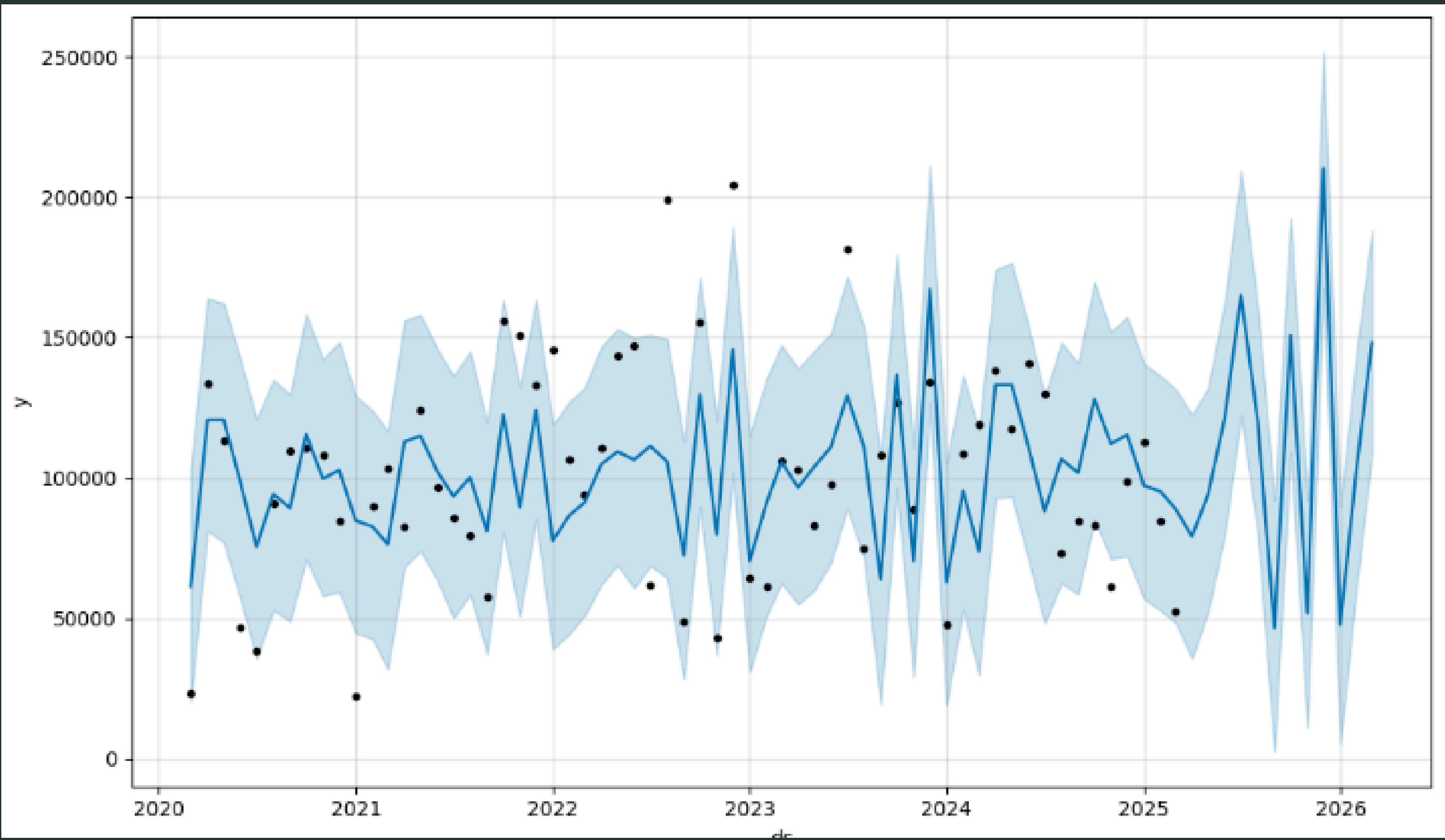


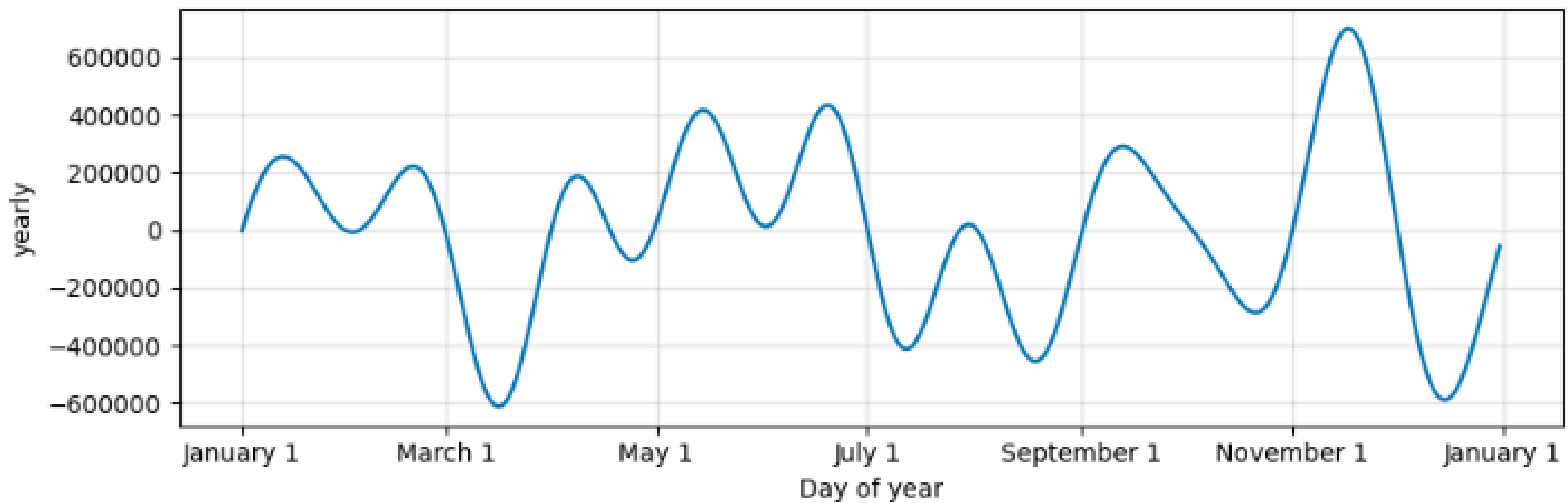
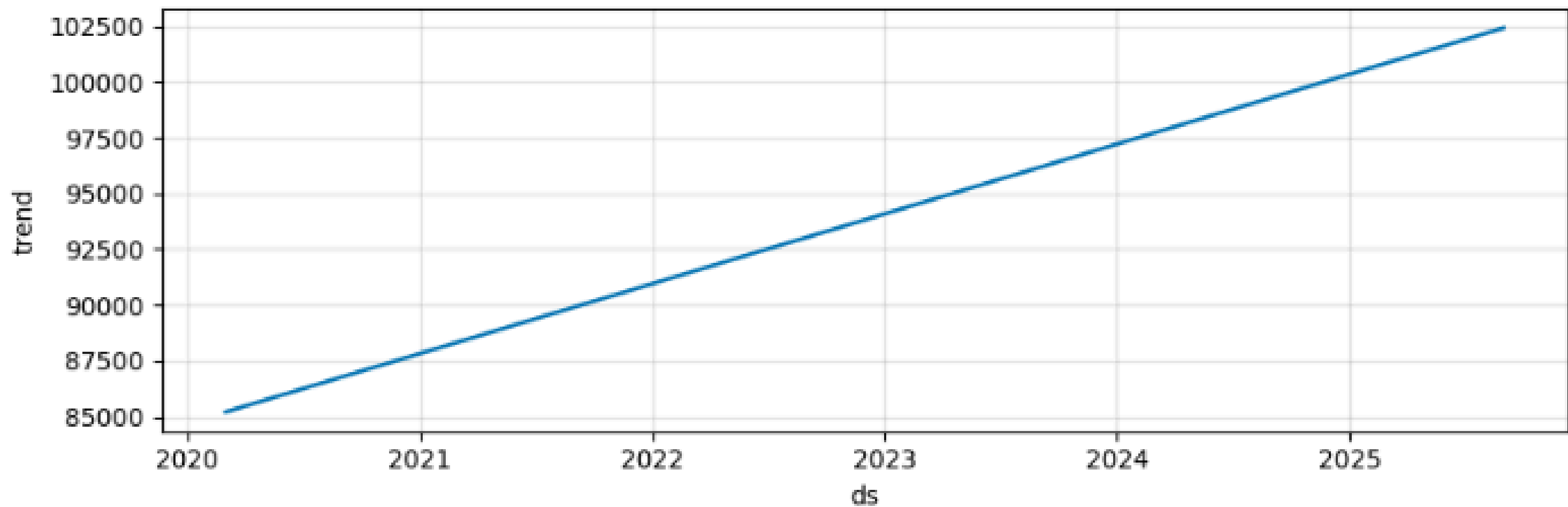
Clusters of Orders (Quantity vs Profit)

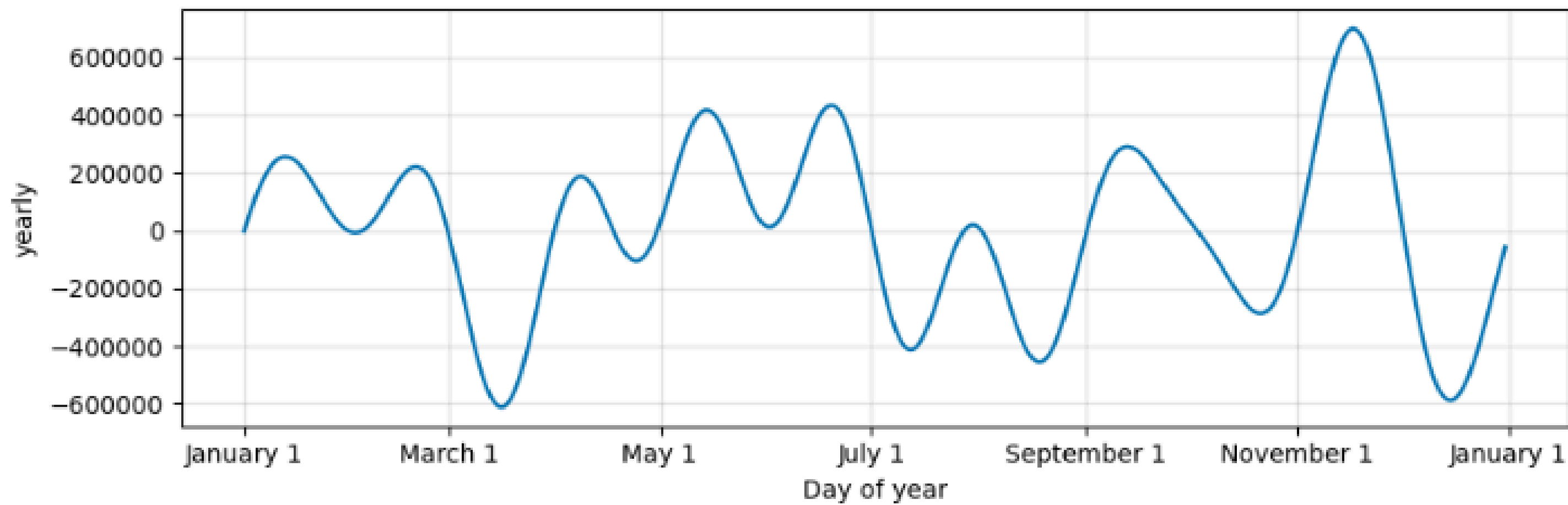
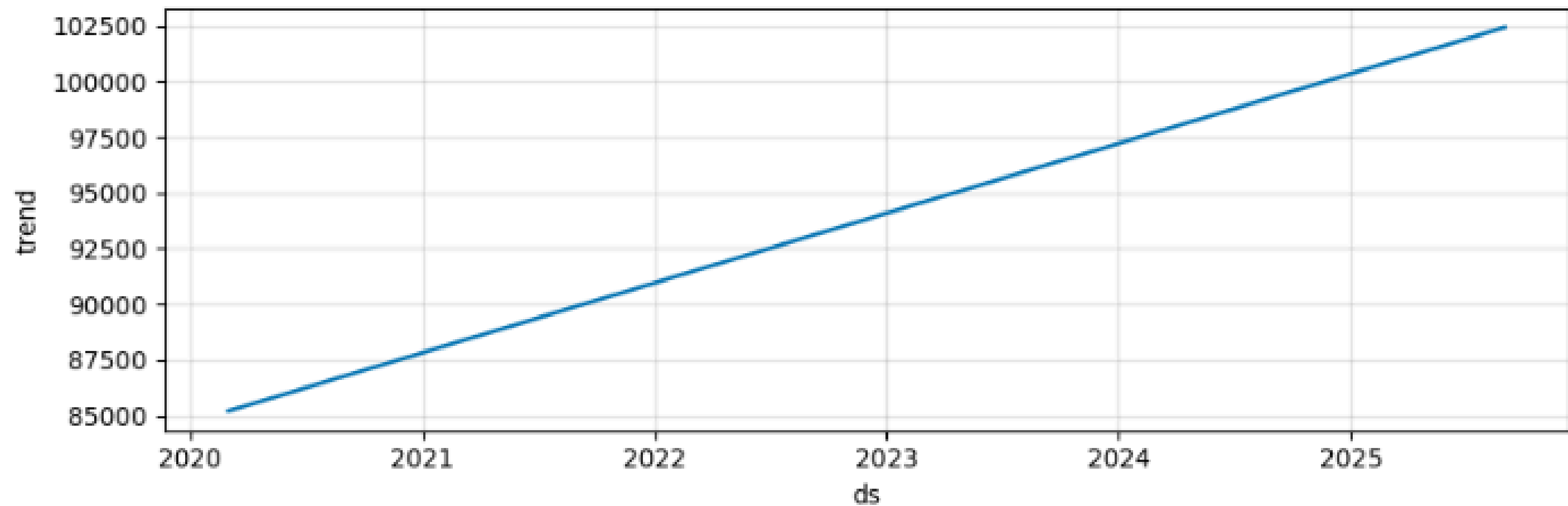


FACEBOOK PROPHET

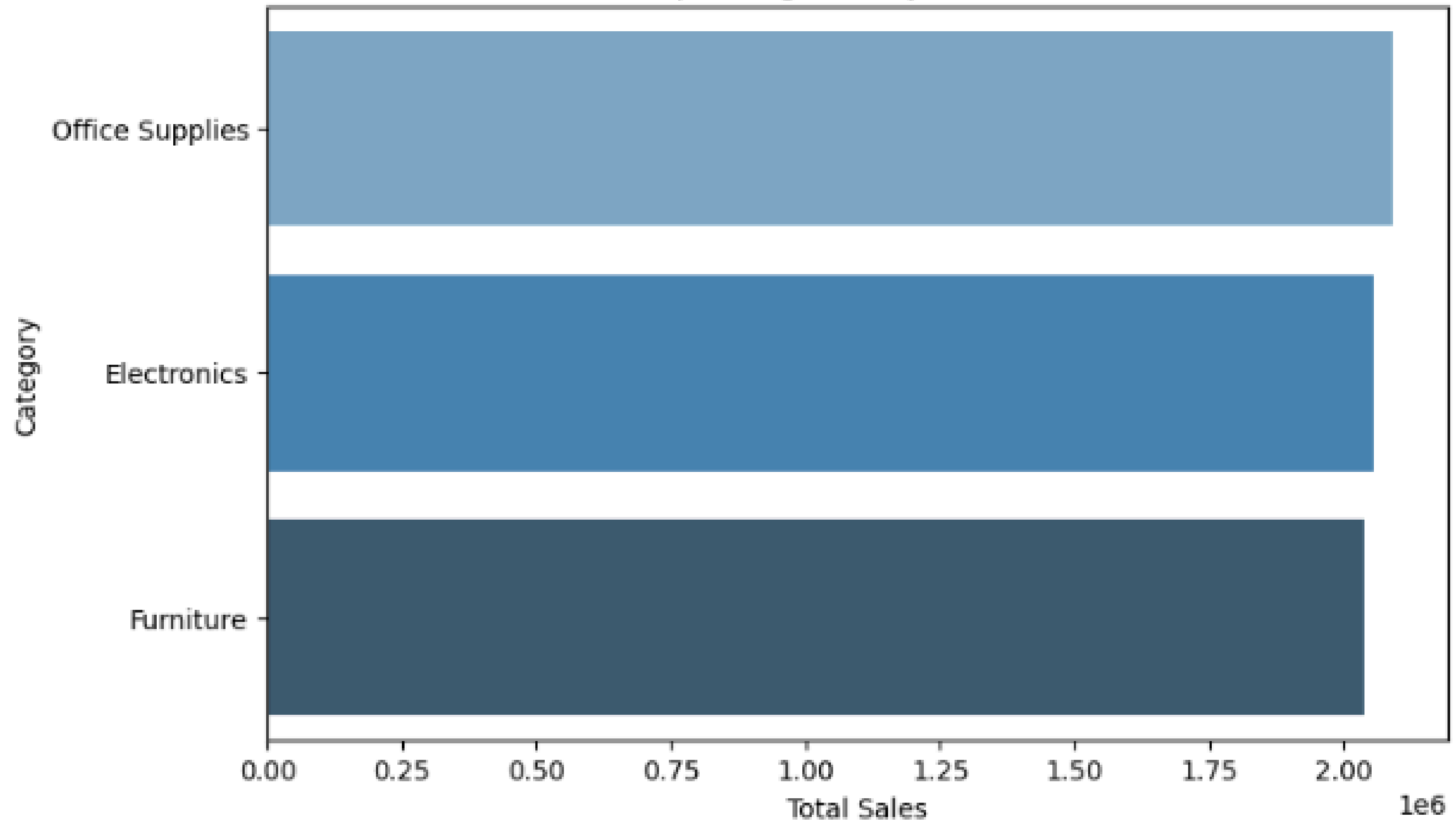
- Facebook Prophet is a time series forecasting algorithm.
- It is specially designed for time series data that shows clear seasonal effects — like sales data, stock prices, website traffic, etc.
- The Prophet model is additive in nature. It breaks down the forecast into components like:
 - Trend – long-term direction of data
 - Seasonality – repeating cycles such as weekends or holidays
 - Holidays/Events – custom or country-specific dates that affect the data
- The user provides the data in a simple format with two columns:
 - ds → the date column
 - y → the numeric value to forecast (e.g., sales, profit, etc.)
- It handles trends, seasonality, holidays, and missing values automatically, making it easy to use even for non-experts.



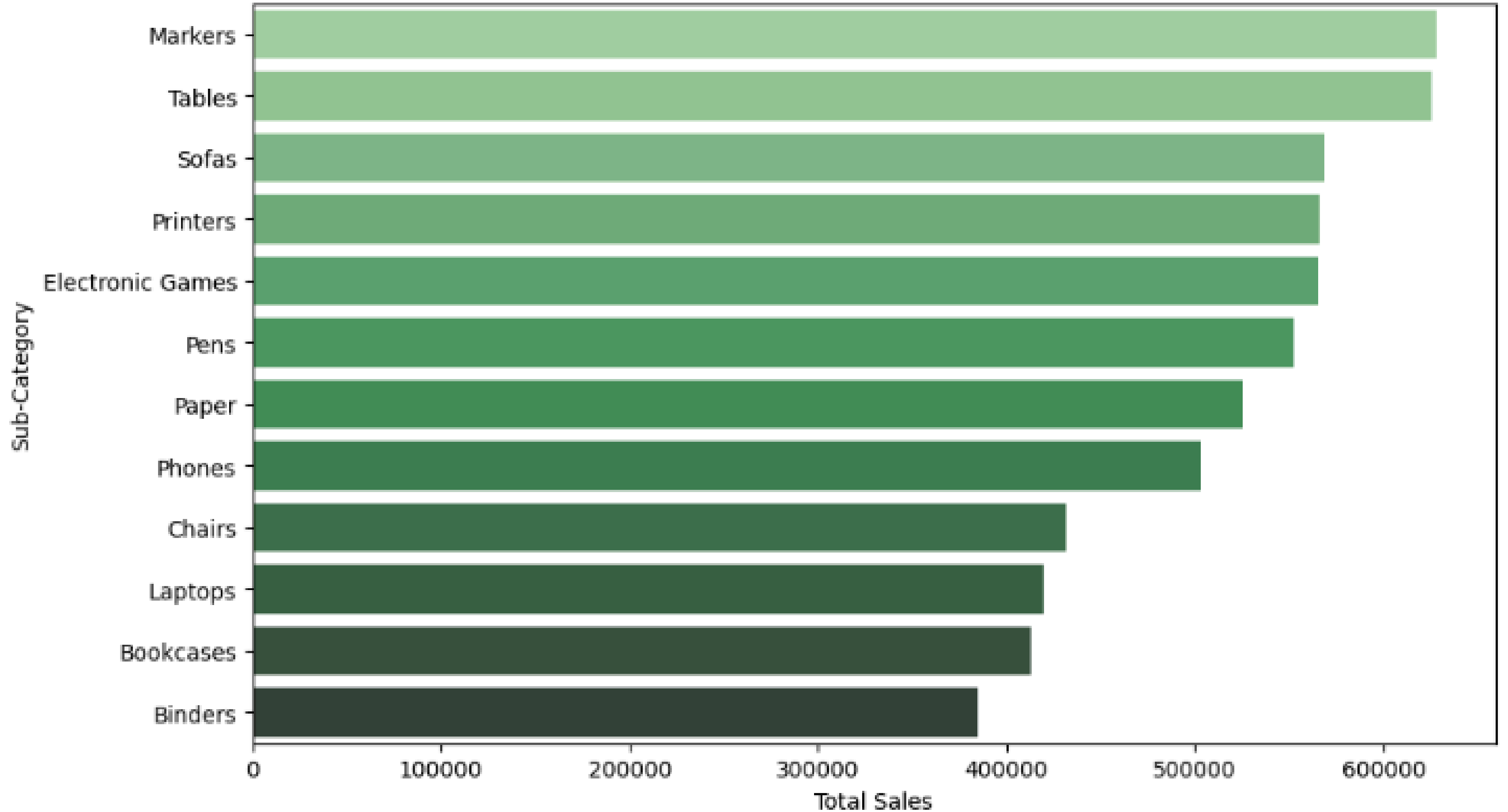




Top Categories by Revenue



Top Sub-Categories by Revenue



Sales and Profit by State

State

New York

Florida

California

Texas

Illinois

Ohio

1130048

1091174

1086436

1011475

978738

884768

Amount

308506

308706

278814

257780

240372

216519

Profit

1e6

1.0

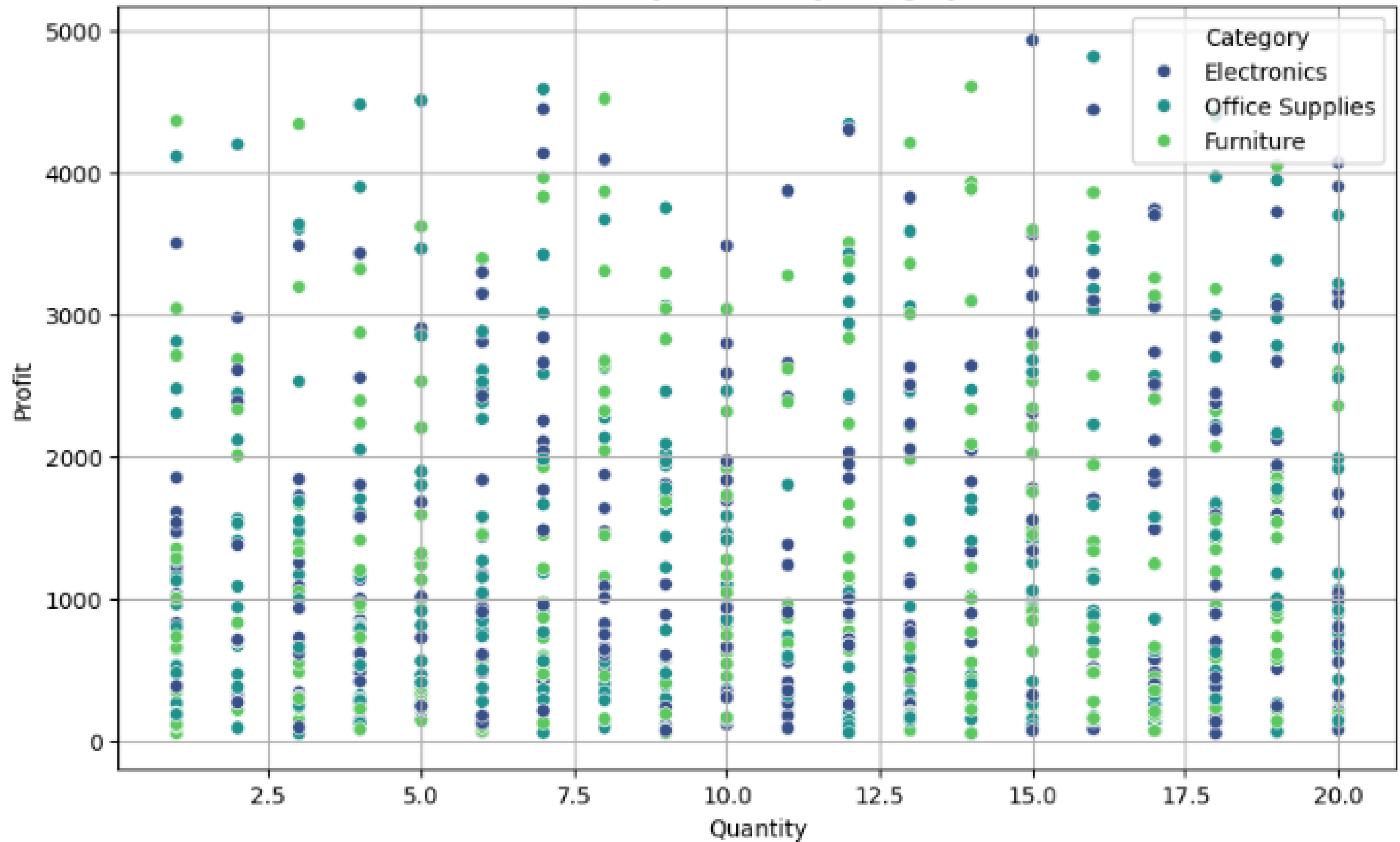
0.8

0.6

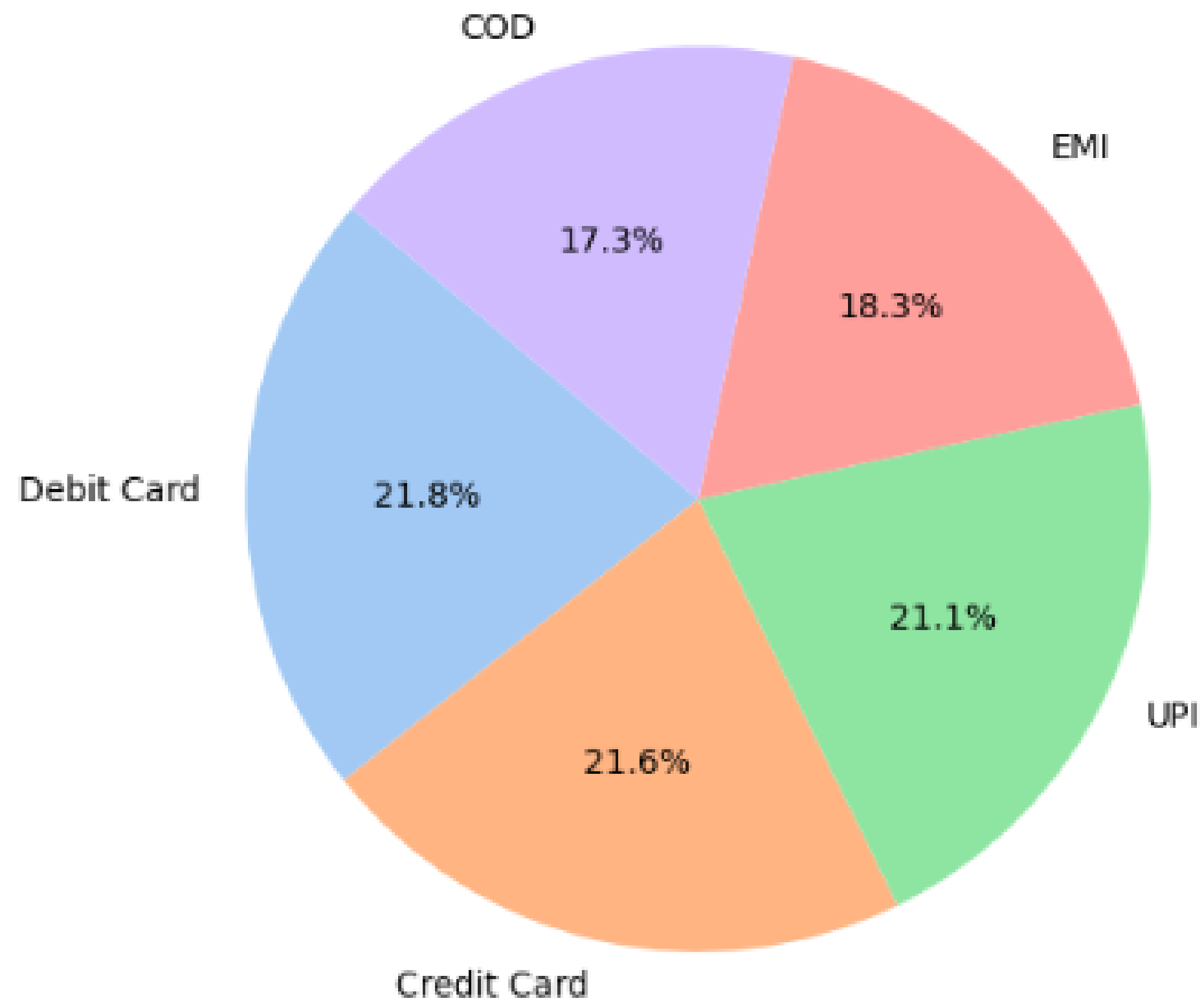
0.4



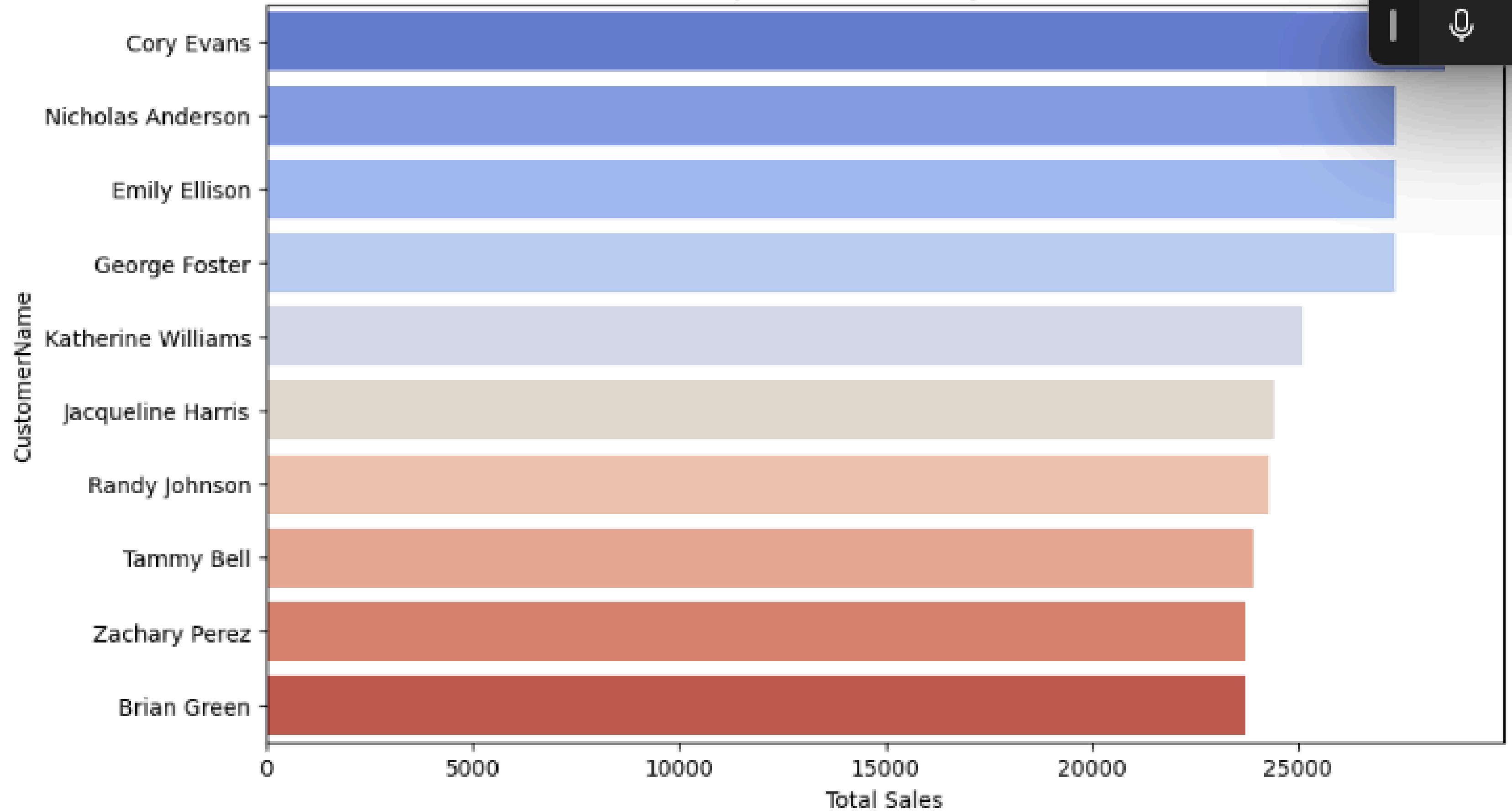
Quantity vs Profit by Category



Payment Method Distribution



Top 10 Customers by Sales



BUSINESS INSIGHTS

- Plan Ahead with Forecasting
Use Prophet to anticipate sales trends and align inventory and staffing.
- Target High-Value Customer
Personalize offers and rewards for top customer segments.
- Expand Geographically
Focus on low-performing states to boost reach and revenue.
- Promote Top Categories
Invest in bundling and cross-selling top products like Phones & Chairs.
- Align Teams with Data Insights
Enable smarter decisions across marketing, ops, and finance

CONCLUSION

The analysis provides a comprehensive view of business performance, highlighting opportunities to optimize inventory, personalize customer engagement, target untapped regions, and improve operational efficiency. Leveraging these insights enables smarter strategies and sustainable growth for the organization.

THANK YOU